Remote Sensing

Rhett D Harrison

Introduction

To do

Fire incidence

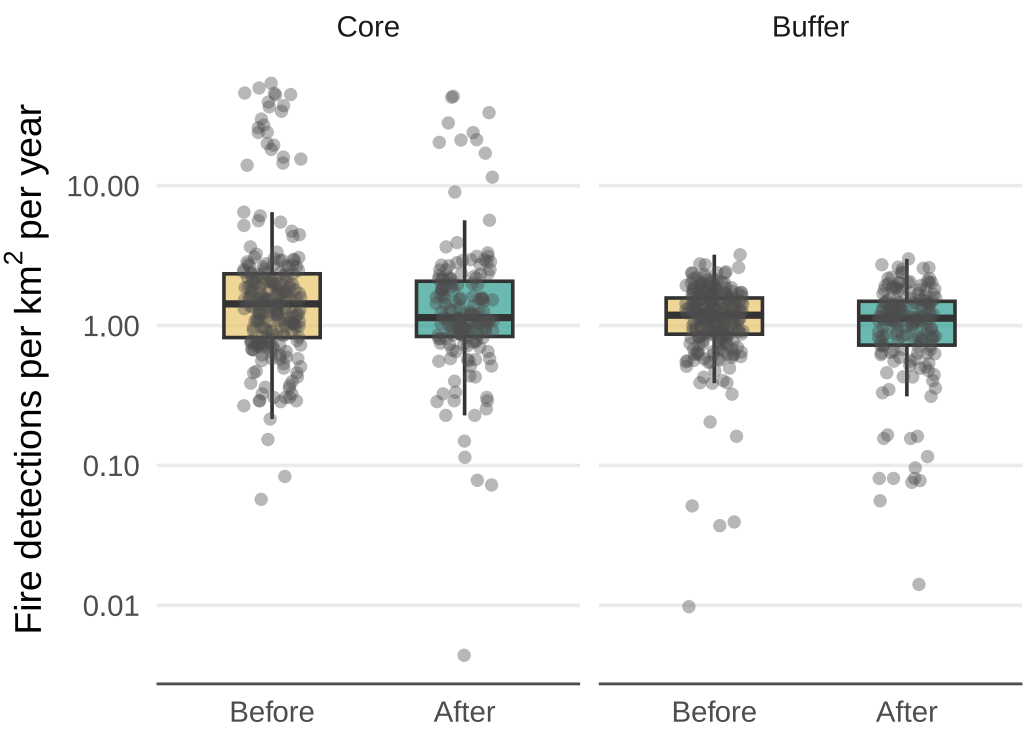

Figure 1: Annual fire detection density (detections per km²) per CFMG before and after establishment, for CF cores and 1 km buffer zones. Each point represents one CFMG-year. Boxes show median and interquartile range.

Table 1: Linear mixed model results for log-transformed annual fire count before and after CF establishment.

Model results for fire detection density (log-transformed) <U+2014> before vs after establishment

Linear mixed model with random intercept for CFMG. Response = log(detections per km<U+00B2> + 1). Fixed effects: period (ref = Before) and governance index. CFMG variance = 0.145; residual variance = 0.191. Marginal R<U+00B2> = 0.055; conditional R<U+00B2> = 0.463. n = 817.

Factor	Estimate	SE	*P* value
Intercept (Before, Core)	1.098	0.161	0.0000
After	−0.121	0.046	0.0083
Buffer	−0.303	0.040	0.0000
Governance index	−0.007	0.030	0.8290
After <U+00D7> Buffer	0.095	0.062	0.1278

Burned area

Table 2: Gamma GLMM results for annual fire footprint fraction before and after CF establishment.

Model results for fire footprint fraction <U+2014> before vs after establishment

Gamma GLMM with log link and random intercept for CFMG. Fixed effects: period (ref = Before) and governance index. CFMG variance = 0.194. Marginal R<U+00B2> = 0.056; conditional R<U+00B2> = 0.437. n = 817.

Factor	Estimate	SE	*P* value
Intercept (Before)	−0.774	0.307	0.0118
After	−0.181	0.068	0.0073
—	−0.318	0.062	0.0000
Governance index	−0.026	0.058	0.6603
—	0.086	0.091	0.3481

Fire radiative power

Table 3: Linear mixed model results for log-transformed annual total FRP per km<U+00B2> before and after CF establishment.

Model results for total FRP per km<U+00B2> (log-transformed) <U+2014> Core vs Buffer, before vs after establishment

Linear mixed model with random intercept for CFMG. Response = log(total FRP / zone area + 1). Fixed effects: period (ref = Before), zone (ref = Core), period <U+00D7> zone interaction (After <U+00D7> Buffer = leakage test), and governance index. CFMG variance = 0.381; residual variance = 0.549. Marginal R<U+00B2> = 0.029; conditional R<U+00B2> = 0.426. n = 817.

Factor	Estimate	SE	*P* value
Intercept (Before, Core)	2.748	0.261	0.0000
After	−0.219	0.078	0.0050
Buffer	−0.298	0.068	0.0000
Governance index	−0.028	0.049	0.5738
After <U+00D7> Buffer	0.110	0.105	0.2991

Deforestation

Table 4: Linear mixed model results for log-transformed annual tree cover loss rate in CF cores and buffer zones before and after establishment.

Model results for annual tree cover loss rate (log-transformed) <U+2014> core vs buffer, before vs after

Linear mixed model with random intercept for CFMG. Response = log(annual loss % + 1). Fixed effects: period (ref = Before), zone (ref = Core), their interaction, and governance index. A significant After <U+00D7> Buffer interaction indicates leakage. CFMG variance = 0.065; residual variance = 0.063. Marginal R<U+00B2> = 0.086; conditional R<U+00B2> = 0.55. n = 803.

Factor	Estimate	SE	*P* value
Intercept (Before, Core)	0.323	0.106	0.0044
After	−0.010	0.026	0.7087
Buffer	0.103	0.024	0.0000
—	−0.015	0.020	0.4737
After <U+00D7> Buffer	0.170	0.036	0.0000

Above ground biomass

Table 5: Linear mixed model results for mean above ground biomass in CF cores and buffer zones before and after establishment.

Model results for mean above ground biomass <U+2014> core vs buffer, before vs after establishment

Linear mixed model with random intercept for CFMG. Fixed effects: period (ref = Before), zone (ref = Core), their interaction, and governance index. A significant After <U+00D7> Buffer interaction indicates leakage. CFMG variance = 363.972; residual variance = 59.557. Marginal R<U+00B2> = 0.052; conditional R<U+00B2> = 0.867. n = 518.

Factor	Estimate	SE	*P* value
Intercept (Before, Core)	49.880	7.815	0.0000
After	4.480	1.087	0.0000
Buffer	−7.974	1.191	0.0000
Governance index	−0.181	1.482	0.9035
After <U+00D7> Buffer	−1.302	1.449	0.3692

Discussion

To do

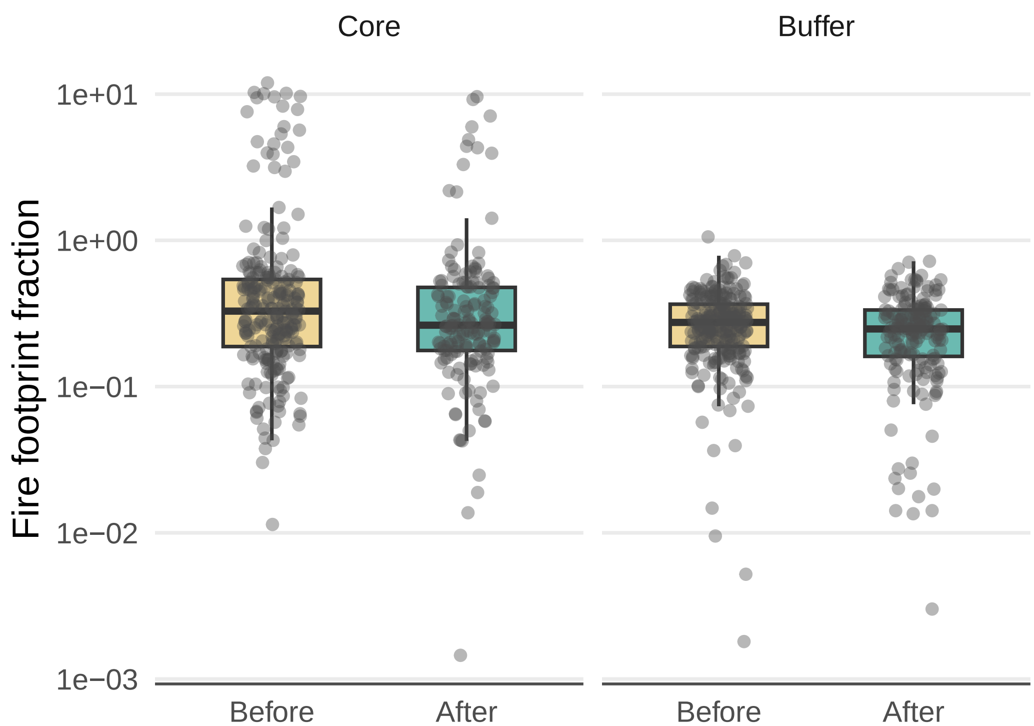

Figure 2: Annual fire footprint as a fraction of area before and after CF establishment, for cores and 1 km buffers. Fire footprint estimated as the sum of scan × track for all detections within each zone-year, divided by zone area. Each point represents one CFMG-year. Boxes show median and interquartile range.

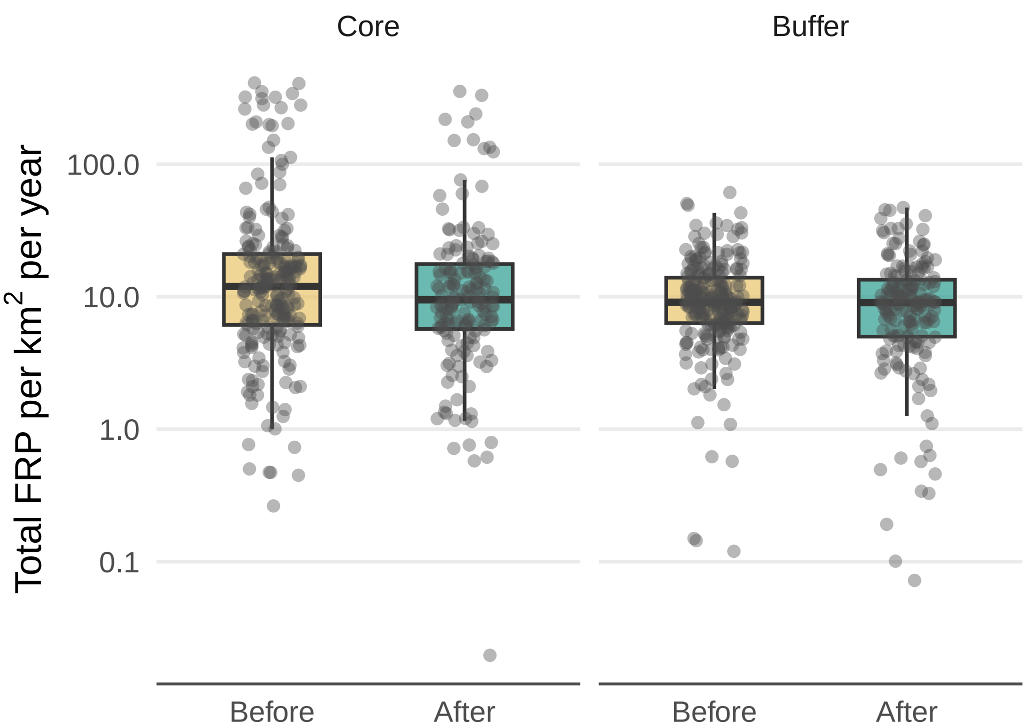

Figure 3: Annual total fire radiative power (FRP) per km² per CFMG before and after establishment, by zone. Total FRP summed across all detections within each CFMG-year-zone and divided by zone area. Each point represents one CFMG-year. Boxes show median and interquartile range. Buffer zone results provide a leakage check.

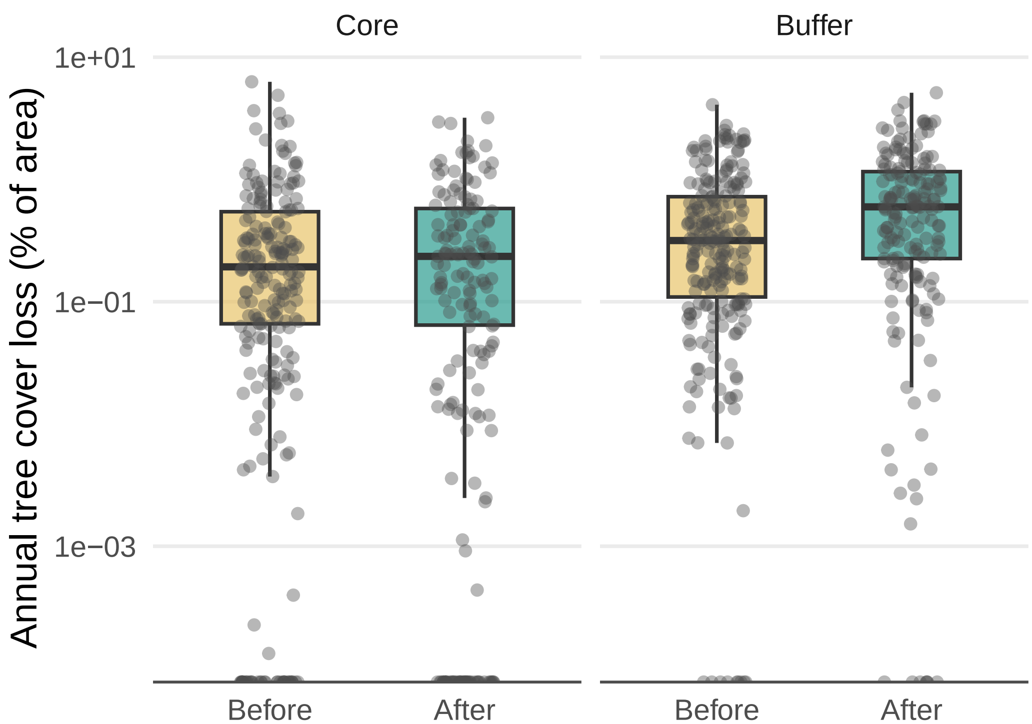

Figure 4: Annual tree cover loss (% of area) in CF cores and 1 km buffer zones before and after CF establishment. Loss derived from Hansen GFC 2024 lossyear data (2014–2024). Each point represents one CFMG-year. Boxes show median and interquartile range.

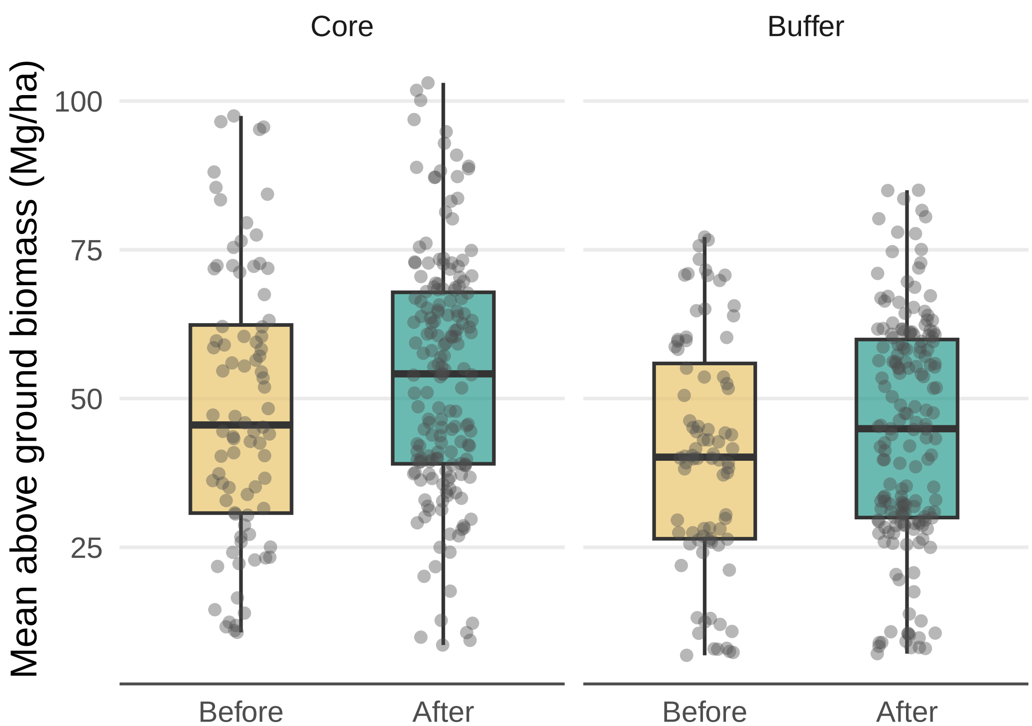

Figure 5: Mean above ground biomass (Mg/ha) per CFMG before and after establishment, for CF cores and 1 km buffer zones. Values are annual means from 2018–2024; core values are area-weighted across polygons. Each point represents one CFMG-year. Boxes show median and interquartile range.